Unit 6, Lesson 9

Recording Partial Products: One-Digit and Three- or Four-Digit Factors

Standards

Addressing **4.NBT.B.5**

Instructional Routines

- Which Three Go Together?

Goals

- Interpret and use an algorithm that uses partial products to multiply multi-digit whole numbers by one-digit numbers.
- Match diagrams and recording methods that represent the same multiplication strategy.

Student Facing Learning Goals

Let's analyze and try an algorithm that uses partial products.

Lesson Purpose

The purpose of this lesson is for students to multiply a multi-digit number by a one-digit number using an algorithm that uses partial products. Students make connections between this algorithm, rectangular diagrams, and equations.

Narrative

In previous lessons, students used diagrams to represent multiplication of a one-digit number and a whole number of up to four digits. Students learned to decompose larger factors by place value and used diagrams and corresponding expressions to support them in finding partial products. In this lesson, students learn an algorithm for keeping track of partial products that come from multiplying the digits of the factors. This algorithm that uses partial products lays the foundation for the standard algorithm for multiplication.

Students engage in quantitative and abstract reasoning (MP2) as they relate the partial products in a diagram and in an algorithm. Because this lesson offers an initial exposure to the new notation, students are not required to use an algorithm that uses partial products to multiply. They can rely on other methods they have learned so far.

Access for Students with Disabilities

- Action and Expression

Access for English Learners

- MLR8

Lesson Timeline

Warm-up	10 min
Activity 1	20 min
Activity 2	15 min
Synthesis Estimate	10 min

Teacher Reflection Questions

What part of the lesson went really well today in terms of students learning? What did you do that made that part go well?

Cool-down 5 min

Warm-up

10 min

Which Three Go Together: Expressions Galore

Standards

Addressing **4.NBT.B.5**

Instructional Routines

- Which Three Go Together?

This *Warm-up* prompts students to carefully analyze and compare four equivalent expressions. In making comparisons, students have a reason to attend closely to the features and the value of the expressions and to use language precisely (MP6). The activity also enables the teacher to gain insight into students' understanding of properties of operations and how they talk about them.

Student Task Statement

Which 3 go together?

A

7×50

B

$(3 \times 50) + (4 \times 50)$

C

$(5 \times 10) \times 7$

D

$50 + 50 + 50 + 50 + 50 + 50 + 50$

Student Response

Sample responses:

- A, B, and C go together because they have a multiplication symbol.
- A, B, and D go together because they include 50.
- A, C, and D go together because each has only one type of operation symbol.
- B, C, and D go together because they have multiple operations.

Launch

- Groups of 2
- Display the expressions.
- "Pick 3 that go together. Be ready to share why they go together."
- 1 minute: quiet think time

Activity

- "Discuss your thinking with your partner."
- 2–3 minutes: partner discussion
- Share and record responses.

Activity Synthesis

- "How are the expressions alike?" (They all have a value of 350.)
- "Were there expressions that you knew right away had the same value? Which ones? How did you know?"

Activity 1

20 min

PLC Activity

An Algorithm for Noah

Standards

Addressing **4.NBT.B.5**

The purpose of this activity is to analyze an algorithm that uses partial products. Students are not required to use a specific notation, but analyzing each algorithm deepens their understanding of the structure of place value in multiplication.

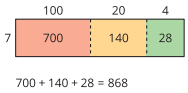
When students interpret and make sense of Noah's work, they construct viable arguments and critique the reasoning of others (MP3).

Access for English Language Learners

MLR8 Discussion Supports. Synthesis: Revoice student ideas to demonstrate and amplify mathematical language use. For example, revoice the student statement: "I added all the numbers together to find the answer" as "I added all the partial products to calculate the final product."
Advances: Speaking, Representing

Student Task Statement

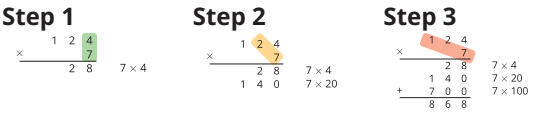
1. Noah draws a diagram and writes expressions to multiply 2 numbers.



7×124
$7 \times (100 + 20 + 4)$
$(7 \times 100) + (7 \times 20) + (7 \times 4)$
$700 + 140 + 28$

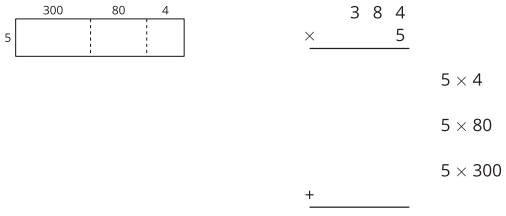
How does each expression represent Noah's diagram?

2. Noah learns another way to record the multiplication.



Make sense of each step of the calculations and record your thoughts.

3. Complete the diagram to find the value of 384×5. Use Noah's recording method to check your work.



Student Response

1. Sample response:
   - 7×124 represents the value of the whole diagram.

Launch

- Groups of 2
- Display the diagram in the activity.
- "What do you notice? What do you wonder?"
- 1 minute: partner discussion
- Share and record responses.

Activity

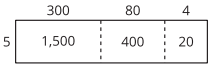
- 4 minutes: independent work time on the first problem about Noah's diagram
- 4 minutes: partner discussion
- 5 minutes: group work time on the rest of the activity
- Monitor for students who include the place value of each digit in 124 in explaining what is happening in the algorithm.

Activity Synthesis

- Select students to share their interpretations of the steps in the second problem.
- Highlight these key ideas along the way:
  - Multiply the 7 by the 4 ones in 124 and write the product below the line.
  - Multiply the 7 by 2 tens in 124 and record this product below the first product.
  - Multiply the 7 by the 1 hundred and record this product below the previous product.
  - Add all the partial products to calculate the final product.

- $7 \times (100 + 20 + 4)$ represents one factor (one side length of the rectangle) being decomposed by place value and being multiplied by 7.
- $(7 \times 100) + (7 \times 20) + (7 \times 4)$ represents each decomposed part of 124 being multiplied by 7 and combined together.
- $700 + 140 + 28$ represents the sum of the products in all the parts, which is the product of 7×124.

2. Sample response: Noah multiplies the single-digit factor by each digit of the multi-digit number at the top in order from right to left. He places the products below the line and adds them up to get the final product.
3. Completed diagram and calculation:



1,500 + 400 + 20 = 1,920

$$\begin{array}{rrrrl} & 3 & 8 & 4 & \\ \times & & & 5 & \\ \hline & & 2 & 0 & 5 \times 4 \\ & 4 & 0 & 0 & 5 \times 80 \\ + & 1, & 5 & 0 & 0 \quad 5 \times 300 \\ \hline & 1, & 9 & 2 & 0 \end{array}$$

$$\begin{array}{rrrr} & 1 & 2 & 4 \\ \times & & & 7 \\ \hline & & 2 & 8 \\ & 1 & 4 & 0 \\ + & 7 & 0 & 0 \\ \hline & 8 & 6 & 8 \end{array}$$

- If needed, review these steps for 5×384.
- Reiterate that we typically begin by multiplying the single-digit factor by the digit in the ones place, and then repeat with the digit in the tens place, and so on.
- "The recording strategy that Noah learned is an algorithm that uses partial products. We used an algorithm to add and subtract large numbers in our last unit."
- "How do we know if we have finished finding all the partial products?" (We keep track of all the partial products or smaller parts of the whole product as we go. We went in order from the smallest place value to the largest place value.)

Activity 2

15 min

Try an Algorithm with Partial Products

Standards

Addressing **4.NBT.B.5**

In this activity, students continue to analyze an algorithm that uses partial products and learn that there are different ways to write the partial products. While students are not required to use a specific algorithm for multiplication, analyzing variations in the partial products notation deepens their understanding of how to use base-ten structure to multiply efficiently (MP7).

Access for Students with Disabilities

Action and Expression: Internalize Executive Functions. Provide access to colored pencils and sticky notes. To support organization, invite students to color code each partial product as seen in the previous activity. To support working memory, invite students to use the sticky notes to record the expanded form of the multi-digit number they are multiplying by.
Supports accessibility for: Memory, Organization

Student Task Statement

Noah and Mai want to find the value of $6 \times 2{,}947$. They record their steps in different ways.

Launch

- Groups of 2
- "What strategy would you use to find the value of

Noah

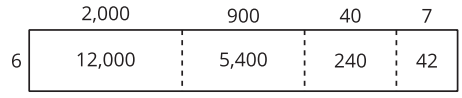
$$\begin{array}{r} 2,947 \\ \times \quad\quad 6 \\ \hline 42 \\ 2,400 \\ 5,400 \\ +\ 12,000 \\ \hline \end{array}$$

Mai

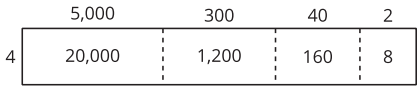
$$\begin{array}{r} 2,947 \\ \times \quad\quad 6 \\ \hline 12,000 \\ 5,400 \\ 2,400 \\ +\ \quad\quad 42 \\ \hline \end{array}$$

1. How are these notations alike? How are they different?
2. Use a diagram to show what each of the partial products 42, 240, 5,400 and 12,000 represent. Then find the value of $6 \times 2{,}947$.
3. Find the value of each expression. Use the algorithm that Noah used for at least one expression. Show your thinking using diagrams, symbols, or other representations.
   a. $4 \times 5{,}342$
   b. 7×983

Student Response

1. They both multiplied using an algorithm and they both used partial products, but Mai multiplied starting with the thousands and Han started with the ones.
2. 

3. a. 21,368. Sample responses:



20,000 + 1,200 + 160 + 8 = 21,368

$$\begin{array}{r} 5,342 \\ \times \quad\quad 4 \\ \hline 8 \\ 160 \\ 1,200 \\ +\ 20,000 \\ \hline 21,368 \end{array}$$

   b. 6,881

- "Take a quiet moment to make sense of Noah and Mai's work. How do you think they arrived at the last four numbers?"
- 1 minute: quiet think time
- 1 minute: partner discussion
- "Which representation is more like how you thought about it?"

Activity

- "Work with your partner on the first two questions."
- 5 minutes: partner work time on the first two problems
- Pause for a brief discussion.
- "How are Mai's and Noah's notation the same and how are they different?" (Both of these methods calculate partial products, but Noah's goes in order from multiplying the ones, tens, hundreds, then the thousands. Mai's goes in order the other way from the thousands, hundreds, tens, and then the ones.)
- "Are both methods correct? How do you know?" (Yes, because they both result in the same product when you calculate it.)
- "Work on the last set of problems independently before discussing with your partner."
- 5 minutes: independent work time on the last set of problems

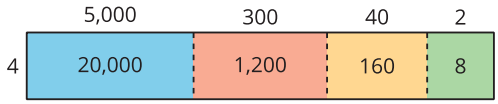
Activity Synthesis

- "In the *Lesson Synthesis*, let's analyze different ways to keep track of partial products."

Lesson Synthesis

"Today we learned different ways of recording partial products to multiply four-digit by one-digit numbers. We made connections between a diagram and using algorithms that use partial products."

Display the expression $4 \times 5{,}342$ and corresponding diagram and computation:



20,000 + 1,200 + 160 + 8 = 21,368

$$\begin{array}{r} 5,342 \\ \times \quad\quad 4 \\ \hline 8 \\ 160 \\ 1,200 \\ +\ 20,000 \\ \hline 21,368 \end{array}$$

Invite students to take turns making connections between each part of the diagram to the algorithm.

Highlighting the following:

- Each number in the smaller rectangles in the diagram is a partial product—a result of multiplying a part of one factor by the second factor.
- When we use a diagram, we can find the partial products for different smaller rectangles and add the pieces together in any order. The product stays the same no matter how we decompose the diagram.
- When we use an algorithm that uses partial products, we multiply the ones, tens, hundreds, then thousands, and record the partial products vertically. Changing the order of multiplying doesn't change the final product.

Suggested Centers

- Number Puzzles: Multiplication (4–5) , Stage 1: Two-Digit Factors (Addressing)
- Compare (1–5), Stage 3: Multiply within 100 (Supporting)

Cool-down

5 min

Partial Products

Standards

Addressing **4.NBT.B.5**

Student Task Statement

Find the value of $5 \times 1{,}023$. Show your thinking using diagrams, symbols, or other representations.

Student Response

5,115. Sample responses:

- $5 \times 3 = 15$, $5 \times 20 = 100$, and $5 \times 1{,}000 = 5{,}000$. The sum of 15, 100, and 5,000 is 5,115.

$$\begin{array}{r} 1,023 \\ \times \quad\quad 5 \\ \hline 15 \\ 100 \\ 0 \\ +\ 5,000 \\ \hline 5,115 \end{array}$$

Responding to Student Thinking

Students decompose factors and multiply without considering place value, resulting in incorrect partial products.

Next Day Supports
After the *Warm-up*, have students discuss with a partner their strategies for the *Cool-down* and allow time to make revisions to their thinking.